

R. Bernadett Vlocskó

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PhD-trained chemist with hands-on organic and a computational chemistry background, seeking a role where I can apply my experience in physics-based and data-driven modeling to address critical challenges in material and process development

- 3+ years leading computational chemistry projects using Gaussian, ORCA, AMS, Schrödinger suites
- 5+ years of hands-on work in synthetic chemistry labs, focusing on reaction optimization and catalysis
- Peer-reviewed publications in high-impact journals including *Journal of Medicinal Chemistry* and *Scientific Reports*
- Vast experience teaching (10 semesters) and mentoring students (2 undergraduate and 4 graduate)
- ~1 year of industry experience delivering projects in cross-functional, interdisciplinary chemistry teams

Education

University of Massachusetts Boston, MA

2020 - Expected May 2026

PhD – Organic Chemistry

Bela Torok's Research [Group](#)

Thesis: Computer-aided development of novel antioxidants

Budapest University of Technology and Economics, Hungary

2017- 2019

Master of Science – Chemical Engineering (Drug Development & Manufacturing)

Functional Organic Materials Research [Group](#), Hungarian Research Network

Thesis: Dynamically controllable interfacial systems containing photoactive azobenzenes

University of Szeged, Hungary

2014 - 2017

Bachelor of Science – Chemistry

MTA-SZTE Stereochemistry Research Group

Thesis: Cinchona alkaloid-catalyzed asymmetric Michael addition of acetylbutyrolactone to activated olefins

Skills

Computational Software & Packages	Quantum mechanics (QM), molecular mechanics (MM), docking, reactivity and mechanism analysis Gaussian 16, ORCA 6.0, BioLuminate (Schrödinger), PyMOL, Quantum ESPRESSO (AMS), COSMOtherm (BIOVIA), Mercury (CCDC), Multiwfn, OriginPro, Mestrelab
Programming	Python, Bash, Unix/Linux
Instrumentation	NMR (¹ H, ¹³ C, ¹⁹ F), UV-Vis, FTIR, GC-QTOF-MS, HPLC-MS, Flash Chrom., XRPD, PLM, TGA, DSC

Experience

PhD research

January 2020 – Present

Computational

- Implemented an integrated MM/QM workflow to evaluate novel small-molecule antioxidants for preeclampsia therapy.
- Benchmarked DFT functionals and basis sets to verify modeling accuracy for diaryl hydrazone derivatives.
- Identified structural drivers of antioxidant activity in diaryl hydrazones and their salt forms to guide lead optimization.
- Validated computational predictions in vitro using human trophoblast cells in collaboration with Beth Israel Medical Center.
- Progressed a lead diaryl hydrazone candidate from in silico design to patent application.
- Identified key physicochemical descriptors governing antioxidant activity in thiophenols via comparative DFT modeling with phenols.
- Clarified solvent-dependent mechanistic pathways (HAT vs. SPLET) to inform structure–activity optimization of antioxidants.
- Built a structure-based docking workflow to assess alcohol dehydrogenase I as a chemoselective biocatalyst using BioLuminate, Epik, Glide, and Prime.

Synthesis

- Developed a scalable, metal- and acid-free protocol for 2-alkyl/aryl benzimidazoles (n = 20) and benzothiazoles (n = 18), achieving quantitative yields under mild conditions. Elucidated reaction kinetics through ¹⁹F NMR reaction profiling.
- Established a high-yielding, additive-free synthesis of diaryl hydrazone quaternary ammonium salts (n = 14), reducing reaction times and simplifying workup.

Agios Pharmaceuticals, Cambridge, MA

June – August 2024

Intern in Drug Substance Group, CMC, R&D

- Integrated the CCDC Database and Mercury software into the in-house salt/cocrystal screening platform.
- Identified, synthesized, and fully characterized two novel salt forms of widely used generic drugs (cimetidine and loratadine).
- Validated simulated single-molecule packing arrangements through XRPD pattern similarity analysis.

Hovione, Lisbon, Portugal

August – December 2019

Intern in Inhalation and Advanced Drug Delivery Group, R&D

- Developed a dry powder liposomal formulation of fluticasone propionate in DSPC for sustained lung delivery.
- Optimized high-pressure homogenization and spray-drying processes to produce reproducible, controlled particle properties.

Master's research

Sept 2017 – July 2019

- Developed controlled interfacial behavior via the isomerization of photoactive azobenzene-based molecular switches.
- Oversaw the synthesis of four molecular switches, and their immobilization on polydopamine (PDA)/Au surfaces.
- Conducted UV-Vis spectroscopy analysis of functionalized surfaces.

Sanofi, Chinoi, Budapest, Hungary

July – August 2018

Intern in Process Safety and Process Control Laboratory, R&D

- Leveraged the CHETAH simulation program for chemical thermodynamic and hazard evaluation.
- Provided group training and authored Hungarian learning materials.

Teaching and mentorship

- Supervised undergraduate (2) and graduate students (5) in experimental design, data interpretation, and laboratory techniques.
- Served as teaching assistant for five years in organic chemistry laboratory courses (188 stud. total).
- Collaborated with multidisciplinary, multicultural teams across chemistry, engineering, and technical operations.
- Demonstrated strong written communication through authoring book chapters, review articles and technical reports.

Presentations

1. **Vločskó, R. B.**, Mastyugin, M., Gustin, L., Zsengellér, Zs., Török, M., Török, B. Exploring Antioxidant Effects of Organofluorine Diphenyl Hydrazone Quaternary Salts in Preeclampsia Therapy: a DFT Approach. *ACS Spring Meeting 2025*.
2. **Vločskó, R. B.**, Mishra, M., Stoica, A., Gustin, L., Török, B. Catalyst-free Approaches in Green Synthesis: Environmentally Benign Synthesis of Benzimidazoles and Benzothiazoles in a Sustainable Solvent. *28th Annual GC&E Conference 2024*.
3. **Vločskó, R. B.**, Mastyugin, M., Török, B., Török, M. Assessing Antioxidant Activity Through Theory and Experiments: A Comparative DFT Study of Phenol and Thiophenol Analogues. *ACS Spring Meeting 2024*.

Peer-reviewed publications

1. Zsengellér, K. Zs., Mastyugin, M., Fusco, R. A., **Vločskó, R. B.**, Ferguson, C., Pintye, D., Nasim, H., Salahuddin, S., Young, C. B., Török, B., Török, M. Impact of Structural Features on the Antioxidant Activity of Organofluorine Diaryl Hydrazones. *Molecules*, **2026**, 31, 78.
2. Mastyugin, M., **Vločskó, R. B.**, Zsengellér, K. Zs., Török, M., Török, B. Development of Diaryl Hydrazones for Alleviation of Mitochondrial Oxidative Stress in Preeclampsia. *Journal of Medicinal Chemistry*, **2025**, 68, 10075-10091.
3. **Vločskó, R. B.**, Mastyugin, M., Török, B., Török, M. Correlation of Physicochemical Properties with Antioxidant Activity in Phenol and Thiophenol Analogues. *Scientific Reports*, **2025**, 15, 73.
4. **Vločskó, R. B.**, Mishra, M., Stoica, A., Gustin, L., Török, B. Catalyst-free Synthesis of Substituted Benzimidazoles and Benzothiazoles in a Sustainable Solvent. *Tetrahedron Green Chemistry*, **2024**, 3, 100035.
5. **Vločskó, R. B.**, Xie, G., Török, B. Green Synthesis of Aromatic Nitrogen-Containing Heterocycles by Catalytic and Non-Traditional Activation Methods. *Molecules* **2023**, 28, 4153.
6. Schäfer, C., Cho, H., **Vločskó, R. B.**, Xie, G., Török, B. Recent Advances in the Green Synthesis of Heterocycles: From Building Blocks to Biologically Active Compounds. *Current Organic Synthesis*, **2022**, 19, 426-462.
7. Kunfi, A., **Vločskó, R. B.**, Keresztes, Z., Mohai, M., Bertóti, I., Ábrahám, Á., Kiss, É., London, G. Photoswitchable Macroscopic Solid Surfaces Based on Azobenzene-Functionalized Polydopamine/Gold Nanoparticle Composite Materials: Formation, Isomerization and Ligand Exchange. *ChemPlusChem*, **2020**, 85, 797.

Awards

Vincent T. Mastricola, Jr. Graduate Fellowship	<i>Top performing grad student, UMB Chemistry Department (2025)</i>
CSM Dean's Doctoral Research Fellowship	<i>UMB College of Science & Mathematics (2025)</i>
Ciba Green Chemistry Travel Award	<i>ACS Green Chemistry Institute (2024)</i>
Graduate Teaching Award	<i>UMB Chemistry Department (2021)</i>
Ivan Villax Scholarship	<i>Hovione, Portugal (2019)</i>
New National Excellence Program Scholarship	<i>National Research and Innovation Office of Hungary (2016 – 2017)</i>

Volunteer experience

Official Notetaker	<i>Computational and Medicinal Chemistry School, Cambridge, MA (2025)</i>
Session Presider	<i>Computational Division, ACS Spring Meeting (2025)</i>
STEM Outreach Mentor	<i>UMass Boston–BCLA-McCormack School partnership (2021 – 2022)</i>